TSA 2015

Problem Solving

Questions taken from the TSA Oxford 2015 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2015 paper, e.g. **1 (Q3)**.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

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1 (Q2) A public telephone has been vandalised and will take only 50 pence coins and 10 pence coins. A woman puts £8.50 into the phone to make a long-distance call with twenty-five coins. Five 10 pence coins are returned when she finishes her call.

What combination of coins is retained by the machine?

A Twice as many 10s as 50s.

B One-and-a-half times as many 50s as 10s.

C The same number of 50s and 10s.

D Three times as many 50s as 10s.

E One-and-a-half times as many 10s as 50s.

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2 (Q6) William wants to cook roast beef and potatoes for himself and four friends. From his cookbook he gets the following information:

BEEF	**Medium oven**	**Hot oven**
Rare	---	10 minutes per 500 g
Medium rare	20 minutes per 500 g plus 20 mins	15 minutes per 500 g plus 15 minutes
Well done	30 minutes per 500 g plus 30 minutes	---
Allow 300 g per person.		

After removing from the oven, the meat should be allowed to stand for 10 minutes.

ROAST POTATOES	**Medium oven**	**Hot oven**
	40 minutes	30 minutes

William buys a joint of the correct weight and decides to cook it medium rare in a hot oven.

How long after putting the beef in the oven should he put in the potatoes, so that both will be ready to serve at the same time?

A 25 minutes

B 30 minutes

C 31 minutes

D 40 minutes

E 50 minutes

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3 (Q7) A teacher is going to award a prize to her most deserving pupil in the current school year. She has narrowed it down to five students and must make her final judgements. She won't give the prize to anyone who has been late to her lessons more than twice. She won't give it to anyone who has failed to complete more than two pieces of homework by the deadline set. After these criteria have been met, the prize will be awarded to the student with the fewest non-A-grade pieces of work.

Pupil	Year group	No. of lessons missed	No. of times late for lesson	No. of homeworks set	Homeworks completed by deadline	Homeworks with grade A
Andrew	8	4	1	60	56	58
Carole	11	2	2	56	56	53
Edward	9	6	2	59	56	58
Grace	10	6	3	58	57	56
Ian	11	10	2	54	53	52

Which student will be awarded the prize?

A Andrew

B Carole

C Edward

D Grace

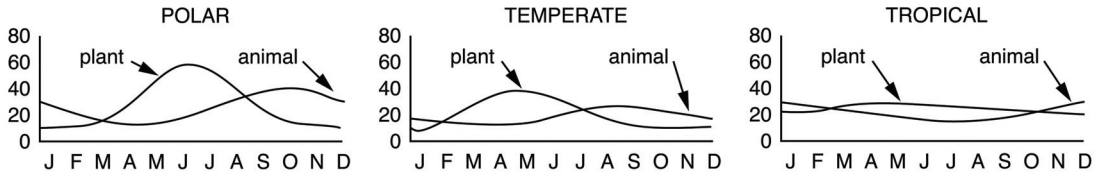
E Ian

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4 (Q8) Plankton are small plants and animals present in the sea. The animal plankton feed on the plant plankton. The rate at which the plant plankton are produced depends on sea temperature and the amount of sunlight.

In tropical regions there is little variation of plankton production through the year. Variations are much bigger at temperate and polar latitudes. Plant plankton production always rises before that of animal plankton.

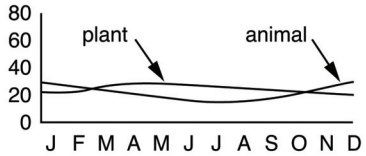
The following graphs show the production rates of plankton in the northern hemisphere:



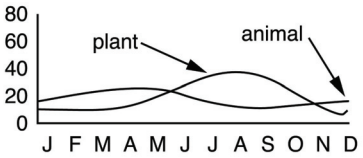
In the northern hemisphere winter is in December, January and February. In the southern hemisphere it is in June, July and August.

Which one of the following graphs provides a good representation of plankton production in the southern polar region?

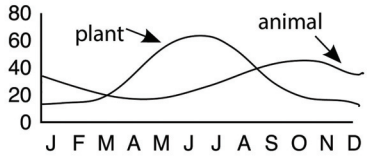
A



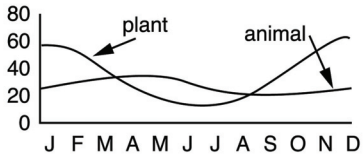
B



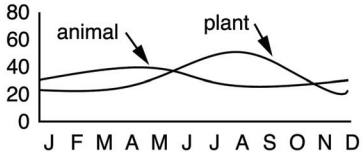
C



D



E



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5 (Q12) A market research company conducted a survey into the relative popularity of two soft drinks with the following results (all figures are percentages):

	Age 5-15	**Age 15-25**	**Age 25-35**	**Age over 35**
Prefer Drink A	28	38	32	24
Prefer Drink B	20	31	33	19
No preference	33	12	22	34
Don't know	19	19	13	23

The same number of people were surveyed in each age group.

Which one of the following conclusions can be justifiably drawn from the results?

A Drink A is consistently more popular than Drink B.

B There were more people with a preference for Drink B than with no preference.

C More people in the age group 5-15 expressed a preference than did not.

D Fewer people had a preference than those who did not.

E More than 20% of people did not know which they preferred.

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6 (Q13) My monthly electricity bill is calculated as a standing charge plus an amount per unit used. At the beginning of one month my electricity company decided to change the tariff by reducing the standing charge and increasing the cost per unit. This was intended to help customers who use less electricity.

The table below shows my consumption and monthly charges for the whole year.

Month	Units used	Charge (£)	Month	Units used	Charge (£)
Jan	125	42.50	Jul	60	36.00
Feb	150	45.00	Aug	70	37.00
Mar	130	43.00	Sep	80	36.00
Apr	110	41.00	Oct	100	40.00
May	95	39.50	Nov	110	42.00
June	75	37.50	Dec	130	46.00

At the beginning of which month was the tariff changed?

A February

B April

C August

D September

E December

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7 (Q14) A survey has shown the following relationships between parental occupation, type of school and educational success at GCSE level:

Parental occupation	Average no. of GCSEs at Grade C or above
Professional	7.3
Managerial	7.2
Routine office work	6.1
Skilled manual	3.3
Semi-skilled manual	2.4
Unskilled manual	1.6

Type of school	Average no. of GCSEs at Grade C or above
Independent (i.e. private)	7.2
Specialist maintained schools	7.0
Local authority comprehensive	4.3
Local authority girls	4.4
Local authority boys	3.8

Which one of the following can be inferred from these tables?

A All children of managerial-class parents attend independent schools.

B The majority of children whose parents are office workers attend independent schools.

C School type is more important than parental occupation in determining educational success.

D The parental background of children in independent schools is more restricted than that of children in comprehensive schools.

E Factors other than type of school attended may be important in determining educational success.

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8 (Q18) There are 420 candidates sitting a public exam in a college. The sports hall can accommodate 210 students; once this has been filled there are other rooms available that are much smaller. The following is a list of the number of single desks that have been set up in each room:

26, 20, 24, 28, 24, 20, 24, 16, 20, 18, 26.

The rooms have sufficient space for all the candidates who are not in the sports hall. The exams officer wants to leave the same number of empty desks in each of these rooms.

What procedure should the exams officer adopt to work out how many students to have in each room?

A Divide the total number of candidates by the total number of workspaces in each room and take that number away from the capacity of each of the rooms.

B Add up the total capacity of the rooms, subtract that number from the total number of candidates, divide the result by the total number of rooms and then subtract that number from the capacity of each room.

C Add up the total capacity of the rooms, subtract the total number of candidates from that figure, divide the result by the total number of rooms and subtract that number from the capacity of each room.

D Divide the total capacity by the number of students and then take that number away from the capacity of each room.

E Add up the total capacity of the rooms and divide by the total number of candidates. Multiply the result by the total number of rooms, which gives the number of students to place in each room.

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9 (Q19) Steve and Jen are getting married and would like to have their reception in a marquee. There will be 80 people attending in total and they would like to have the most expensive tables and chairs – eight people will sit at each table. The capacity for a marquee can be worked out using the industry standard of allowing 3 m^2 per person. Price lists from the hire company they will use are given below:

Size (m)	Size (ft)	Marquee	Carpet	Lighting	Lining	Complete Package
6 x 3	20 x 10	£150	£45	£25	£80	£300
6 x 6	20 x 20	£200	£60	£35	£95	£390
6 x 9	20 x 30	£235	£85	£55	£115	£490
6 x 12	20 x 40	£280	£95	£45	£120	£540
6 x 15	20 x 50	£380	£125	£80	£155	£740
6 x 18	20 x 60	£440	£145	£90	£185	£860
9 x 3	30 x 10	£175	£65	£45	£95	£380
9 x 6	30 x 20	£235	£85	£55	£115	£490
9 x 9	30 x 30	£320	£105	£65	£135	£625
9 x 12	30 x 40	£380	£125	£75	£155	£735
9 x 15	30 x 50	£440	£145	£80	£185	£850
9 x 18	30 x 60	£520	£165	£90	£230	£1,005
9 x 21	30 x 70	£590	£185	£100	£260	£1,135
9 x 24	30 x 80	£680	£205	£110	£280	£1,275
9 x 27	30 x 90	£730	£225	£120	£310	£1,385
9 x 30	30 x 100	£810	£245	£130	£380	£1,565

Item	Cost
Bistro chair	£1.30
Gilt chair	£2.40
Trestle table (seats 6)	£4.00
Round table (seats 8)	£6.00
Patio table (seats 4)	£4.00

What is the lowest total cost of a complete marquee package plus tables and chairs for Steve and Jen's reception?

A £982

B £1,385

C £1,529

D £1,637

E £1,817

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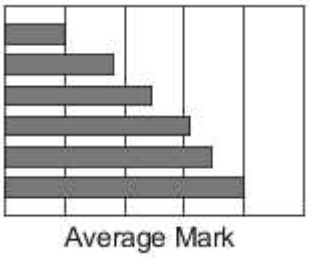
10 (Q20) The maths sets at Newtown Comprehensive are identified by their teacher's name. Every year the Head of Maths gives the same test to all the maths sets at the end of year 7, calculates the average mark for each set and then produces the results on a bar chart in ascending order. He is asked to present the graphs for the last five years to the Board of Governors. Unfortunately, when they were cut and pasted onto one sheet by the Clerk to the Governors, the years were missed off.

The results for last year were as follows:

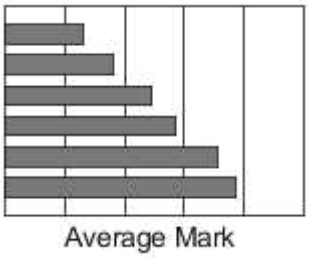
Set	7BR	7EV	7LE	7MO	7RA	7SP
Av. mark	57	36	82	20	71	49

Which one of these five graphs shows last year's results?

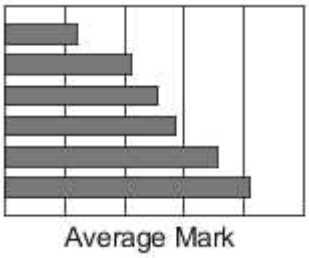
A



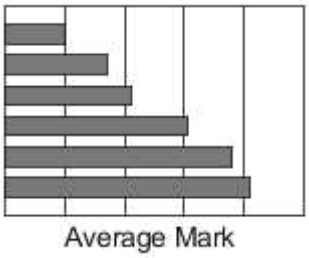
B



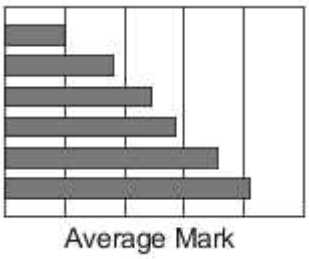
C



D



E



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11 (Q24) Nowadays many people in the UK work out their own income tax bills. Everyone has a personal allowance which is the amount of money they can earn each tax year before they start paying tax. Income above the personal allowance is taxed in bands. The UK tax bands and personal allowances for four tax years are shown below. The tax year runs from 6 April to 5 April the following year.

TAX RATES		**1999-2000**	**2000-01**	**2001-02**	**2002-03**
Starting Rate	at 10% on the first	1500	1520	1880	1920
Basic Rate	at 22% on the next	----	26880	27520	27980
	at 23% on the next	26500	----	----	----
Higher Rate	at 40%	over 28000	over 28400	over 29400	over 29900
Personal Allowance					
Aged under 65		4335	4385	4535	4615
Aged 65 and over but under 75		5720	5790	5990	6100
Aged 75 and over		5980	6050	6260	6370
Married Couple's Allowance					
Elder spouse aged					
Under 65		1970	----	----	----
65 and over but under 75		5125	5185	5365	5465
75 and over		5195	5255	5435	5535

Mary is 52 and her husband is 58. Her only earnings each year are from a summer job. Between 6 April 2000 and 5 April 2001 she earned £5,585.

How much tax did she pay?

A None

B £120

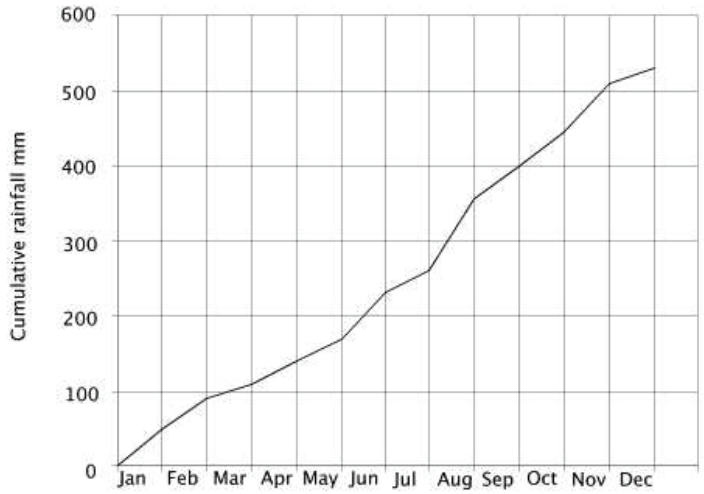
C £125

D £152

E £1,200

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12 (Q25) The graph below shows the cumulative rainfall for Malvern Wells in the UK for 2010.



What, approximately, was the average monthly rainfall from the **beginning** of June to the **end** of September?

A 30 mm/month

B 40 mm/month

C 48 mm/month

D 57 mm/month

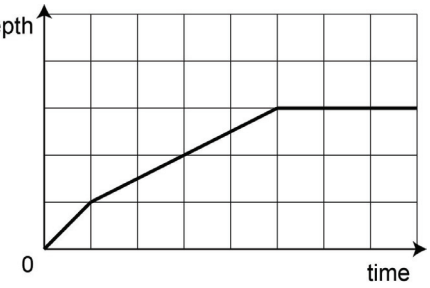
E 100 mm/month

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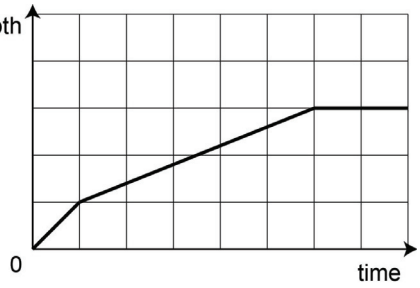
13 (Q26) Duncan's bath has a flat base and vertical sides. It can be filled completely from the hot tap in 15 minutes or from the cold tap in 10 minutes. Its capacity is 360 litres. When preparing a bath, Duncan's habit is to run both taps together for 1½ minutes, before turning off the cold tap. He turns the hot tap off when the bath is ¾ full, then leaves the water for a while to cool down to a suitable temperature before he climbs in.

Which graph would show how the depth of water increases as Duncan prepares his bath?

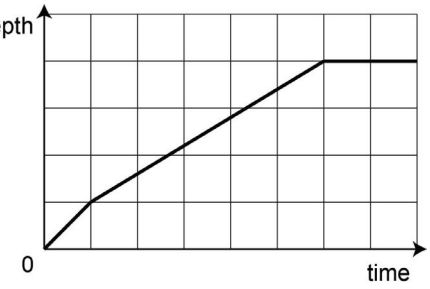
A



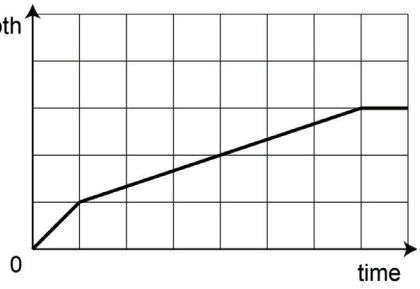
B



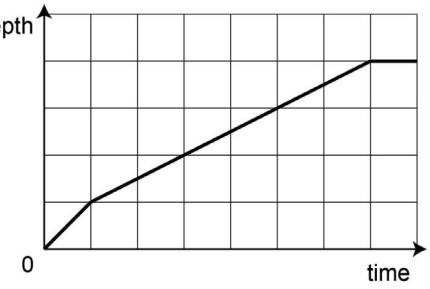
C



D



E



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14 (Q30) Towards the end of a sports season in which all teams in the league play each other four times, the league positions are as follows:

	Played	**Won**	**Drawn**	**Lost**	**Pts**
East Rovers	16	8	7	1	31
West Athletic	14	7	3	4	24
North Rangers	17	6	4	7	22
South United	17	4	10	3	22
Top Town	13	3	8	2	17
Bottom Albion	15	3	6	6	15

Three points are awarded for a win, 1 point per draw and 0 for a loss. At the end of the season, if two teams have the same number of points, the league will be decided by the highest number of games won.

How many of the competing teams could win the league when all the games have been completed?

A 1

B 2

C 3

D 4

E 5

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15 (Q31) I am planning to take up a new hobby. There are a number of classes available locally, summarised in the table below:

[I do not want to have to leave home before 09:15 and I must be back by 16:15. I will choose the activity that is the cheapest per hour.]

Activity	Travel time (one way)	Start time	Finish time	Class price ($)
Painting	20 mins	10:00	12:00	30
	20 mins	10:30	13:00	40
	50 mins	10:00	12:00	24
Dancing	30 mins	9:30	13:00	45
	10 mins	13:00	17:00	45
Pottery	60 mins	10:00	11:30	35
	60 mins	11:00	13:00	40
Writing	10 mins	10:00	12:30	35
	30 mins	14:00	15:30	24

What will be the price of the class I will take?

A $24

B $30

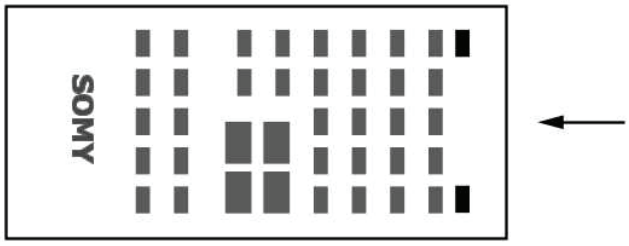
C $35

D $40

E $45

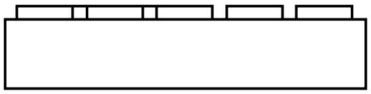
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16 (Q32) My TV remote control is shown below. All the buttons stick out from the body of the remote control.

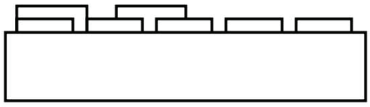


When viewed from the direction of the arrow, which one of the diagrams below is **not** a possible view of the remote control?

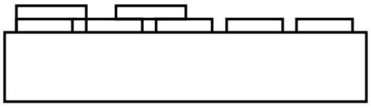
A



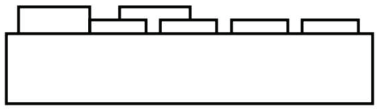
B



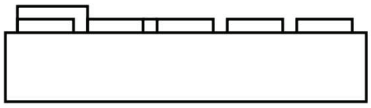
C



D



E



17 (Q36) A film star has agreed to sign autographs for £1 each at a charity function, all day from 9 am to 6 pm.

He claims to sign his name consistently ten times every minute, but insists on having a ten-minute break after each full hour of signing.

How much do the organisers hope that the autographs will raise?

A £4,500

B £4,600

C £4,700

D £4,800

E £5,400

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18 (Q37) A gardener has to apply mulch to two plots of land, measuring 10 metres by 5 metres and 15 metres by 20 metres. The mulch has to be applied to a depth of 50 mm for the first plot and 100 mm for the second plot. The mulch weighs 0.65 tonnes per cubic metre.

	AREA COVERAGE OF MULCH IN SQUARE METRES					
Quantity in cubic metres	**25 mm depth**	**50 mm depth**	**75 mm depth**	**100 mm depth**	**125 mm depth**	**150 mm depth**
1	40	20	13	10	8	7
2	80	40	26	20	16	14
3	120	60	39	30	24	21
4	160	80	52	40	32	28
5	200	100	65	50	40	35
6	240	120	78	60	48	42
7	280	140	91	70	56	49
8	320	160	104	80	64	56
9	360	180	117	90	72	63
10	400	200	130	100	80	70

How many one-tonne bags are necessary to complete the mulching of the two plots?

A 2

B 13

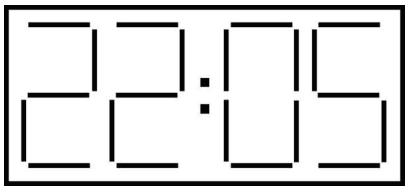
C 20

D 22

E 33

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19 (Q38) I stayed in a hotel last night. My room was at the front of the building. At one point I looked out of the window and saw:

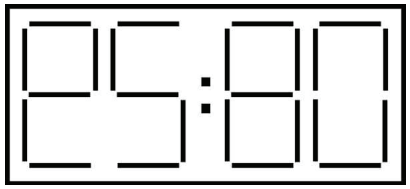


on an illuminated digital clock on the other side of the road. I took the opportunity to reset my watch, which had stopped sometime earlier.

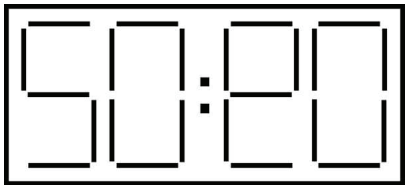
This morning, when I looked out, I realised that what I could see across the road was a reflection in the building opposite of a clock outside the hotel. According to my watch (which I now knew I had reset incorrectly last night) the time was supposedly nine minutes past ten.

What did I see across the road when I looked out this morning?

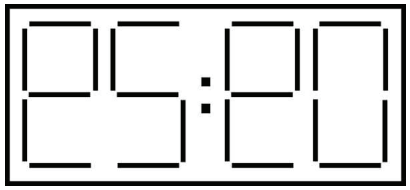
A



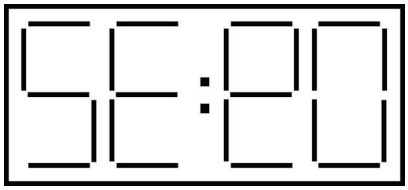
B



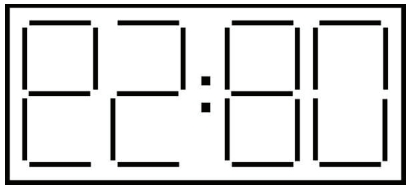
C



D



E



20 (Q42) An examination contains five questions. Candidates are instructed to answer three questions only, including at least one of the first two and no more than one of the last two.

How many different combinations of questions are available for a candidate to choose from?

A 4

B 6

C 7

D 8

E 10

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21 (Q43) Hester is going for a job interview. She needs to travel by railway between Birmingham and Derby, but she needs to stop off in Tamworth for an hour on the way out to see her former teacher for some coaching. When at Derby, she needs an hour and a half to go to the prospective employer to take an aptitude test. She can fix her appointments for any time in the day. Naturally, she wants to make the journey in the most efficient way so that it takes the least possible time.

This is the railway timetable:

Out:

Station								
Birmingham	09:00	10:00	11:00	12:00	13:00	13:30	14:30	15:30
Tamworth	09:30	10:28	11:30	12:45	-	13:45	14:45	15:50
Derby	09:50	10:40	11:50	13:05	13:35	14:00	15:05	16:00

Return:

Station								
Derby	10:40	11:10	12:15	13:10	14:00	15:20	16:10	17:00
Tamworth	11:05	-	12:35	13:45	-	15:45	16:35	-
Birmingham	11:50	11:55	13:10	14:15	15:00	16:12	16:55	18:15

Which would be the best trains to catch from Birmingham and Derby?

A Leaving Birmingham 09:00; leaving Derby 12:15

B Leaving Birmingham 10:00; leaving Derby 14:00

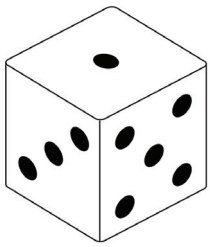
C Leaving Birmingham 11:00; leaving Derby 15:20

D Leaving Birmingham 12:00; leaving Derby 16:10

E Leaving Birmingham 13:30; leaving Derby 17:00

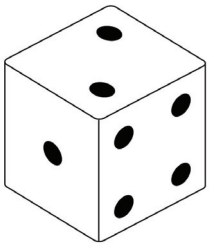
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22 (Q44) The die shown below is a conventional die, i.e. the total number of spots on any two opposite faces is always seven.

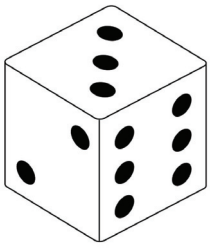


Which one of the following could **not** be the appearance of the die shown above after a subsequent throw?

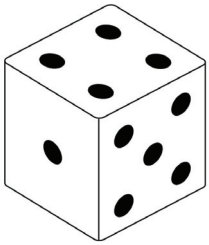
A



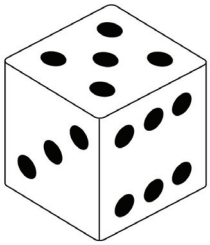
B



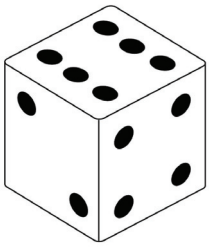
C



D

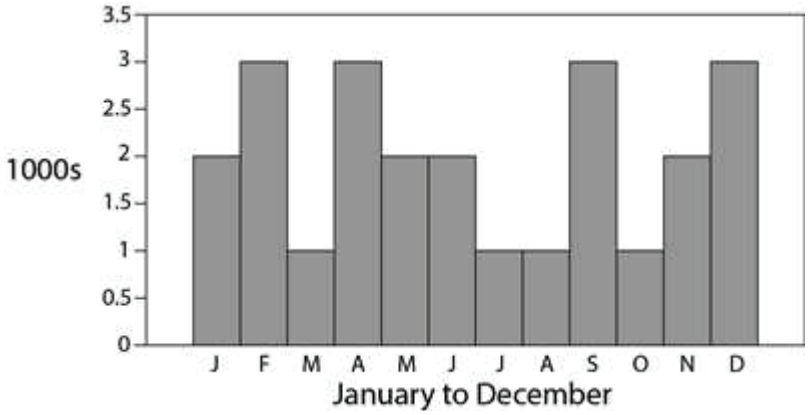


E



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23 (Q48) Deliveries of containers for a fast-food takeaway are made only four times per year, at the beginning of January, April, July and October. Last year the takeaway over-ordered containers and currently has 2,000 left. The number of containers used per month varies, but is no more than 3,000 and no fewer than 1,000. The maximum number that can be delivered at one time is 6,000. The usage for last year is illustrated below.



Assuming the same usage this year, how many containers need to be delivered each quarter (in January, April, July and October) in order for over-ordering to be minimised?

A 5,000 in January, 6,000 in April, 5,000 in July, 6,000 in October

B 6,000 in January, 7,000 in April, 5,000 in July, 6,000 in October

C 4,000 in January, 7,000 in April, 5,000 in July, 6,000 in October

D 6,000 in January, 6,000 in April, 6,000 in July, 6,000 in October

E 4,000 in January, 6,000 in April, 6,000 in July, 6,000 in October

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24 (Q49) The table below shows the available denominations of British stamps in 2009:

Stamp	Value	Usual Purpose
1p	1p	Make up value
2p	2p	Make up value
5p	5p	Make up value
9p	9p	Make up value
10p	10p	Make up value
17p	17p	Make up value
20p	20p	Make up value
22p	22p	Make up value
2nd	30p	2nd Class (UK) standard letter rate
1st	39p	1st Class (UK) standard letter rate
Large 2nd	47p	2nd Class (UK) standard letter rate
50p	50p	Make up value
54p	54p	Worldwide basic (20 g) surface rate
56p	56p	European basic (20 g)
Large 1st	61p	1st Class (UK) standard letter rate
Postcard	62p	World postcard rate
62p	62p	Airmail

What is the lowest amount above 55p which must be made up with at least three stamps?

A 58p

B 65p

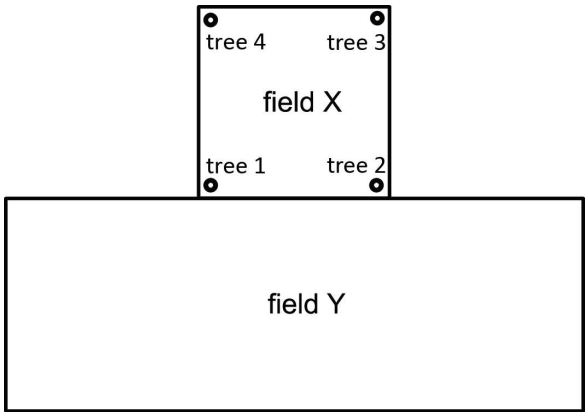
C 68p

D 75p

E 79p

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25 (Q50) Four trees stand at the corners of square field X. They are photographed from field Y, from every possible direction, close up and from a distance, but never so that one tree stands in front of another.



In how many different orders, left to right, can the trees appear in the photographs?

A 1

B 2

C 3

D 4

E 5

END OF TEST

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